Assignment Report: Velocity-Time and Distance-Time Analysis

1. Objective

The objective of this assignment is to analyze vehicle motion using velocity-time data collected from a recorded journey and to compute the distance traveled using numerical methods. The calculated distance is then compared with the actual distance obtained from Google Maps.

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2. Data Collection

The journey was recorded using Google Maps for a duration of 20 minutes covering more than 10 km. Velocity readings were manually recorded at every 30-second interval, resulting in multiple data points representing real driving conditions.

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3. Methodology

3.1 Velocity-Time Analysis

The collected velocity data was plotted against time to understand the variation of vehicle speed during the journey.

3.2 Distance Calculation

Since velocity varies with time, the total distance cannot be calculated using simple multiplication. Therefore, numerical integration was used.

The trapezoidal rule was applied:

$d = \Sigma\ [(v_1 + v_2)/2 \times \Delta t]$

Where:

- v_1 and v_2 are consecutive velocity values
- Δt is the time interval (30 seconds)

3.3 Distance-Time Graph

The cumulative distance was calculated and plotted against time to observe how distance increases over time.

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4. Results

- Calculated Distance = 6525 meters (6.525 km)

The velocity-time graph shows fluctuations due to real traffic conditions such as braking, acceleration, and congestion.

The distance-time graph shows a non-linear increasing trend due to varying velocity.

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5. Error Calculation

Error was calculated using:

Error = |Actual Distance − Calculated Distance|

Percentage Error = (Error / Actual Distance) × 100

Example:

- Actual Distance = 6800 meters
- Calculated Distance = 6525 meters

Error = 275 meters
Percentage Error = 4.04%

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6. Observations

- Velocity fluctuates significantly due to traffic conditions.
- Smaller time intervals improve accuracy.
- Manual recording introduces minor errors.

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7. Conclusion

The trapezoidal numerical integration method provides a good approximation of the total distance traveled. The calculated distance closely matches the actual distance, with a small percentage error. Automation using Python can improve accuracy and reduce manual effort.

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8. Future Work

The process can be automated using video processing and OCR techniques to extract velocity directly from recorded videos.